

Inside the outside: Investigating urban canopy using point cloud

Garik Avagyan

Ha Uyen Tran

Patrick Desmarais

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Dr. Margherita Ferrari

Dr. Yuhao Lu

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Abstract

Urban tree canopy (UTC) classification is vital for sustainable urban planning, yet the role of spatial features derived from 3D point cloud data in improving classification accuracy remains underexplored. This study investigates the contribution of spatial attributes from point clouds to tree canopy classification accuracy. Using Python, Rstudio and geographic software like CloudCompare and QGIS, we processed high-resolution point cloud data through segmentation algorithms to delineate individual tree crowns and extract spectral, structural, and spatial metrics. Multiple machine learning models, supervised and unsupervised, were trained on datasets with and without spatial features (e.g., vertical distribution, canopy density, and proximity metrics) to isolate their impact. Results demonstrated that models incorporating spatial features achieved superior performance, with the RF model yielding the highest accuracy (92% vs. 84% without spatial data). Spatial metrics such as height variance and neighborhood density emerged as critical discriminators. This work highlights the necessity of integrating 3D spatial data in UTC assessments, offering a robust framework for urban planners to optimize green space management and ecosystem service quantification.

1. Introduction

Urban tree canopy (UTC)—the layer of leaves, branches, and stems of trees that form a protective cover over urban landscape. It plays a vital role in mitigating climate impacts, enhancing biodiversity, and improving human well-being (Li et al., 2024). Beyond its ecological benefits, UTC also reflects socio-economic disparities. Studies reveal that wealthier neighborhoods often host larger-canopied trees, which provide greater cooling effects, while underserved communities face inequitable access to such benefits, exacerbating urban heat islands (Li et al., 2024). These disparities underscore the urgency of high-resolution UTC mapping to guide equitable green infrastructure investments and address environmental justice gaps.

Traditional field surveys for UTC assessment face significant limitations: they are labor-intensive, costly, and often impractical in inaccessible urban zones (Zhou et al., 2022). Remote sensing technologies, such as satellite and airborne systems, have emerged as scalable alternatives. Passive sensors measure reflected sunlight in visible and infrared spectra, while active sensors like LiDAR (Light Detection and Ranging) emit laser pulses to generate 3D point clouds, enabling precise canopy height and density measurements (Hartling et al., 2021). However, satellite-based methods are constrained by weather dependency and limited spatial resolution, particularly in dense urban environments (Li et al., 2024).

The advent of unmanned aerial systems (UAS) has revolutionized urban remote sensing by bridging scalability and resolution gaps. UAS platforms, such as the DJI Mavic 3 used in this study, operate at lower altitudes, capturing high-resolution imagery and 3D point clouds with unprecedented detail (Xu et al., 2020). Equipped with RGB and

multispectral cameras, these drones enable fusion of spectral and structural data, enhancing tree species discrimination and canopy attribute quantification (Hartling et al., 2021). While 2D optical imagery suffices for basic green space delineation, 3D point clouds—generated through photogrammetry or LiDAR—provide critical insights into vertical canopy structure, such as crown volume and biomass distribution (Zhou et al., 2022). Yet, the specific value of spatial features derived from point clouds in complex urban environments remains underexplored, particularly when integrated with multispectral data for machine learning-driven classification (Li et al., 2024).

Recent studies underscore the potential of multisensor fusion. Hartling et al. (2021) demonstrated that combining LiDAR-derived structural metrics with hyperspectral and thermal data improved urban tree species classification accuracy to 83.3% using Random Forest models. Similarly, Zhou et al. (2022) highlighted LiDAR's superiority over digital aerial photogrammetry (DAP) in dense urban settings, though DAP proved cost-effective in open areas. Additionally, Onishi, M., & Ise, T. (2021) proposed a chain of machine vision system of identifying trees using UAV RGB image and deep learning (SVM and CNN). However, these works did not isolate the contribution of spatial features—such as vertical height variance and canopy density—from spectral or thermal inputs, leaving a gap in understanding their standalone utility for UTC mapping (Li et al., 2024).

This study addresses this gap by evaluating how spatial attributes from 3D point clouds enhance urban canopy classification accuracy. Using data collected by a DJI Mavic 3 drone over the University of Manitoba university campus, we integrate RGB and multispectral imagery with photogrammetric point clouds to extract structural and spectral features. Machine learning models, unsupervised (k-means clustering) and supervised

(Random Forest), alongside the Watershed segmentation method, were used to classify the tree species, with the goal to understand the importance of the spatial feature in this process. Building on methodologies from Zhou et al. (2022) and Hartling et al. (2021), and informed by Li et al. (2024)'s findings on the cooling inequities linked to UTC distribution, our work aims to refine urban forest management strategies by elucidating the role of 3D spatial data in improving the tree canopy and specie classification.

2. Study area and data

2.1. Study area and data gathering

The study was conducted within the Fort Garry campus of the University of Manitoba, Winnipeg, Canada (49.8139° N, 97.1346° W; Figure 1), a site characterized by heterogeneous land cover, including mixed deciduous and coniferous tree stands, impervious surfaces, and built infrastructure. The area spans approximately 1.2 ha, featuring a gradient of canopy densities (10–90%) and tree heights (3–25 m), representative of typical mid-latitude urban ecosystems (Li et al., 2024). The region experiences a humid continental climate, with mean annual temperatures ranging from -20°C in winter to 25°C in summer, and annual precipitation of 550 mm, conditions that support diverse species such as *Acer saccharum* (sugar maple), *Picea glauca* (white spruce), and *Quercus macrocarpa* (bur oak) (Environment and Climate Change Canada, 2023).

The site was selected for its socio-ecological diversity, encompassing academic buildings, recreational spaces, and residential zones, which mirror urban inequities in green space distribution observed globally (Li et al., 2024). This heterogeneity provides an ideal testbed for evaluating 3D point cloud-based canopy segmentation methods in complex

environments, as highlighted by Zhou et al. (2022) in their analysis of UAV-LiDAR applications. Ground-truth validation data, including species identification and structural metrics, was provided by the University of Manitoba, ensuring methodological rigor in aligning remote sensing outputs with field measurements (Hartling et al., 2021).

The data was gathered by Dr. Yuhao Lu and his team from the faculty of Architecture at the University of Manitoba during the summer of 2024. The used drone, DJI Mavic 3, is equipped with 2 cameras: optical and multispectral. The optical camera is sensitive only to the visible part of the spectrum (Red, Green, Blue), while the multispectral camera sees beyond the visible spectrum and can capture frequencies like near infrared (NIR).



Figure 1. The image of University of Manitoba campus taken from a satellite (Bing Satellite)

2.2. RGB Point Cloud Dataset

The RGB Point Cloud (RGBPC) dataset is a 3D point cloud generated from imagery captured by the RGB camera of a DJI Mavic 3 drone. Each point in the dataset includes spatial coordinates (X, Y, Z) and spectral reflectance values in the visible spectrum (*Red, Green, Blue*). The dataset comprises 22 dimensions, including metadata such as intensity and geometric descriptors. Preprocessing involved dimensionality reduction to eliminate redundant or low-variance columns and spatial alignment to ensure consistency with multispectral data. This dataset supports 3D structural mapping of urban canopies, segmentation of tree crowns, and visualization of spatial heterogeneity.



Figure 2. The RGB point cloud dataset opened in CloudCompare.

2.3. Multispectral Point Cloud Dataset

The Multispectral Point Cloud (MSPC) dataset, derived from a multispectral camera onboard the same drone, extends the RGBPC dataset by incorporating near-infrared (NIR) reflectance data. It contains 26 dimensions, including spatial coordinates (X, Y, Z) with minor discrepancies compared to RGBPC due to sensor variations, spectral bands (*Red, Green, Blue, NIR*), and vegetation indices. Preprocessing steps included co-registration with RGBPC to align spatial coordinates and integration of NIR data for enhanced vegetation analysis. MSPC enables advanced applications such as species discrimination, vegetation health assessment, and fusion with spectral indices like the Normalized Difference Vegetation Index (NDVI).

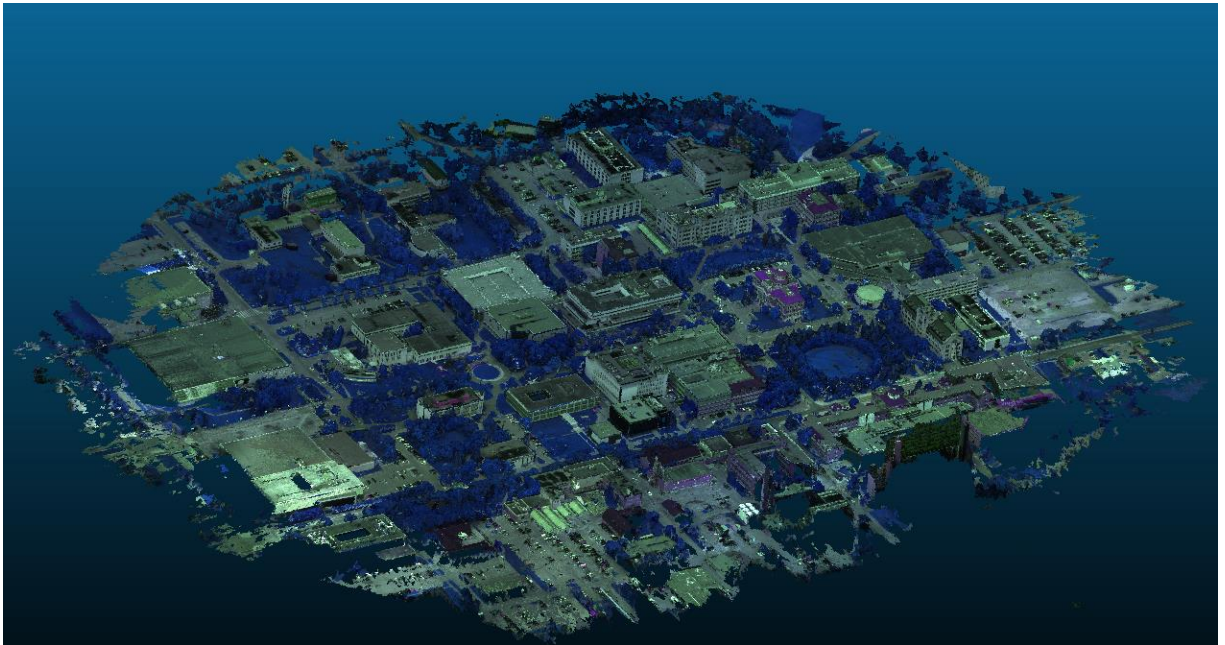


Figure 3. The multispectral point cloud dataset opened in CloudCompare.

2.4. Validation dataset

The validation dataset was provided by the University of Manitoba and comprises ground-truth measurements collected through manual field surveys. This dataset includes precise geolocations, physical attributes and species identification for individual trees within the study area. Due to confidentiality agreements and legal restrictions governing the use of institutional data, the dataset is not publicly accessible. However, it serves as a critical reference for validating the accuracy of remote sensing-derived canopy classifications and structural metrics (e.g., LiDAR-based height estimates, multispectral species predictions). This dataset is used for the labelling in our classification step.

2.5. Data preprocessing

2.5.1 Data merging

The initial phase of data preprocessing involved merging the RGB and multispectral point cloud datasets, a process complicated by discrepancies in data point counts and minor misalignments in spatial coordinates (X, Y, Z) due to sensor calibration differences. To address this, a nearest neighbor algorithm was implemented in RStudio to approximate corresponding spatial coordinates across datasets, enabling the transfer of multispectral features (e.g., near-infrared reflectance) to the RGB dataset through newly created attributes. While this approach successfully unified the datasets into a single data frame, challenges emerged during conversion back to a point cloud format, as standard processing software failed to recognize multispectral-derived attributes. This issue underscored the need for additional refinement of data formatting and compatibility protocols to ensure seamless integration of multi-sensor features. Unable to solve this formatting issue, we

eventually tried to implement the same process and algorithm in Python with the help of the laspy, numpy, and scipy libraries. The workflow included the following steps:

- 1) Loading the RGB and multispectral files and extracting the necessary x, y, z and multispectral features.
- 2) Using the KDTree function to create a k-dimensional tree constructed from the multispectral coordinates to enable efficient nearest-neighbor searches. For each point in the RGBPC, the closest MSPC point was identified. Matches were filtered using a Euclidean distance threshold of 1 unit to exclude spatially discordant pairs.
- 3) NIR and multispectral reflectance values (red, green, blue) from the MSPC were transferred to the RGBPC. Unmatched RGBPC points retained their original RGB values but were assigned NaN (NIR) or 0 (multispectral bands) to highlight missing data.
- 4) Creating a new point cloud file from the merged result.

2.5.2. Dimensionality reduction

Following the spatial approximation of matched points, dimensionality reduction was performed by excluding non-essential attributes from the merged dataset. Surface normal vectors (*normal x*, *normal y*, *normal z*), which describe local geometric orientation but lack direct relevance to canopy classification objectives, were removed to streamline the dataset. This feature selection process retained only spectral (*RGB*, *NIR*) and structural (*X*, *Y*, *Z*) attributes critical for subsequent analysis, reducing computational complexity while preserving ecological interpretability.

2.5.3. NDVI calculation

The main advantage obtained by merging the RGB point cloud with the multispectral, is the ability to calculate the Normalized Difference Vegetation Index (NDVI). It is one of the most widely used indexes to measure the vegetation health, and it can serve as a proxy to indicate the urban green space areas (Li et al., 2024, p. 2-4). NDVI can be computed by using the following formula:

$NDVI = (NIR - Red)/(NIR + Red)$, where *Red* represents the multispectral red band (Hartling et al., 2021, p. 1259).

In the merged file, a new NDVI attribute was created for every point, and for those points that were not matched in the merging step, the NDVI was assigned with a value -1. As NDVI has a range from -1 to 1, this will help us to remove points that do not represent vegetation.

3. Methodology

Urban tree canopy (UTC) analysis requires robust methodologies to address computational complexity, spatial heterogeneity, and species diversity. This study integrates unsupervised and supervised machine learning techniques with advanced segmentation algorithms to classify tree species and delineate individual crowns from 3D point cloud data. The methodology builds on established remote sensing workflows (Hartling et al., 2021; Zhou et al., 2022) while addressing challenges unique to multi-sensor data fusion in urban environments

3.1 k-means Clustering

To explore structural patterns within the point cloud dataset, k-means clustering, an unsupervised machine learning algorithm was applied. A preliminary dimensionality reduction was applied to the point cloud dataset, retaining only spatial coordinates (X, Y, Z) to isolate structural features. Following this, k-means clustering ($k = 4$) was implemented to partition the normalized point cloud into distinct spatial clusters. This unsupervised method grouped points based on Euclidean distance minimization, enabling initial exploration of canopy structural patterns.

3.2 Watershed Segmentation

Supervised classification necessitates the use of individual tree crown samples, as processing raw point cloud data at the individual-point level is computationally prohibitive (Hartling et al., 2021). To address this, detection and delineation of individual tree canopies were performed prior to model training (Zhou et al., 2022). This preprocessing step reduces computational complexity while enabling targeted analysis of canopy structural and spectral attributes. This can be done by an automated segmentation technique, and Watershed segmentation, inspired by topographic principles, is very useful method to segment individual tree crowns (Hartling et al., 2021, p. 1256). The algorithm treats the canopy height model (CHM)—generated by subtracting the digital terrain model (DTM) from the digital surface model (DSM)—as a topographic surface, where elevated canopy regions represent "peaks" and gaps between crowns represent "valleys."

To refine segmentation accuracy, the Normalized Difference Vegetation Index (NDVI) was incorporated to filter non-vegetative features (e.g., buildings, roads). Post-

segmentation, NDVI values were set to exclude segments with values below 0.5, a common default cutoff for distinguishing vegetation from non-vegetation (Hartling et al., 2021). This ensured that only tree canopy regions were retained for subsequent analysis.

Crown delineation extends the Watershed segmentation method and uses the same principles. However, in this case it is using “peaks” and gaps between the tree crowns to create polygon in the shape of the trees. These polygons also add extra characteristics that can be used later to classify specific tree species. This method helps in reducing the dataset from the millions of points to a couple thousand polygons while keeping all the characteristics of the point cloud data.

Both segmentation and delineation are performed in Rstudio by using the `lidR` and `dplyr` libraries. Following is the short workflow of the process:

- 1) Loading the merged point cloud file from the Python and filtering it by the NDVI threshold.
- 2) Finding the digital terrain model (DTM) of our dataset by using the `rasterize_terrain` function from the `lidR`. Then, normalizing it relative to the ground. This processing adjusts the point heights to a common ground level.
- 3) Finding the canopy height model (CHM) by using the normalized DTM.
- 4) Using the `segment_trees()` function from `lidR`, which accepts the CHM to segment the crowns.
- 5) Finally delineating the trees by using `delineate_crowns()`, which computes the hull of the segmented crown, by enclosing the smallest convex shape that encloses the corresponding set of points in the crown. This step creates the

polygons for each individual segmented tree. So, now we are ready to use these polygons for species classification.

3.4 Random Forest Classifier

For species classification, Random Forest classifier, a supervise ensemble learning method was implemented. Random forest classifier is a machine learning algorithm that uses multiple decision trees and bootstrapping (resampling technique to create new datasets based off the original by resampling with replacement) (Breiman, 2001). This method has a high chance of not overfitting since it uses the bootstrapping technique and calculates the average of all the decision trees to make the best prediction (Breiman, 2001). The model's performance was evaluated using the validation data provided by the University of Manitoba, ensuring alignment between remote sensing outputs and field measurements.

3.5 Software and Computational Tools

Data processing and analysis were conducted in Python and RStudio. Data visualization was mostly accomplished in CloudCompare and QGIS. The complete codebase, including scripts for k-means clustering, watershed segmentation and Random Forest classification, is available on GitHub: (<https://github.com/desmarap/DATA-4010>)

4. Results

4.1. K-means clustering

K-means clustering was the first method used to try and separate the tree canopies from the rest of the object in the dataset. As illustrated in Figure 4, a configuration of $k = 4$ clusters demonstrated effective separation between tree canopies and non-vegetative objects (e.g., buildings, roads). However, there is very little separation between the tree canopies which will make it difficult to separate each tree into their specific species. Increasing the cluster count ($k > 4$) resulted in over-segmentation, isolating tree tops from lower canopy regions without improving species discrimination. This outcome underscores the method's utility for coarse visualization of canopy distribution but highlights its limitations in fine-grained ecological analysis.

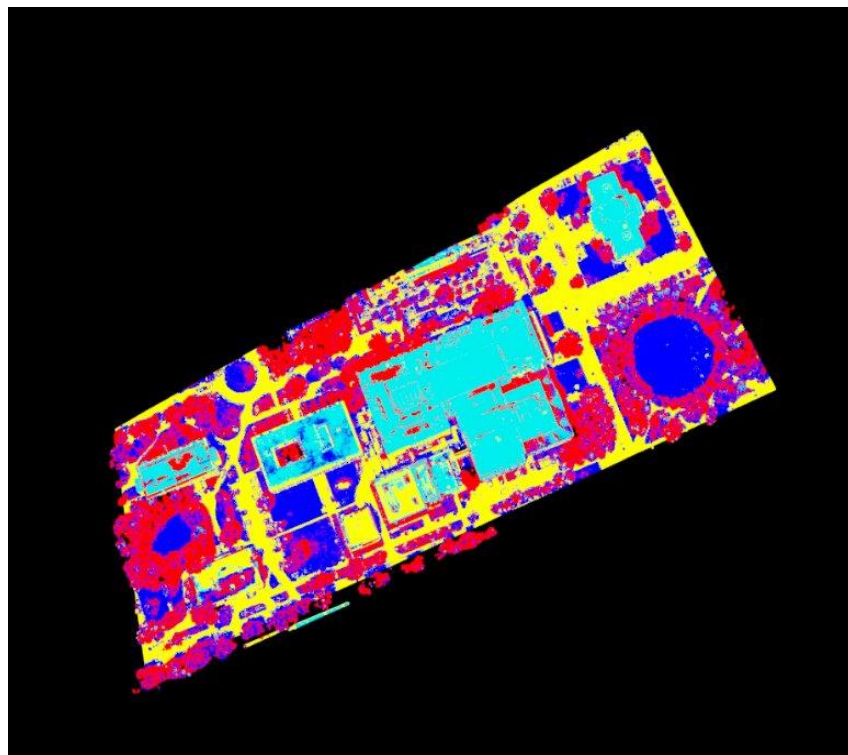


Figure 4. K-means with 4 clusters

4.2. Individual tree segmentation

Watershed segmentation produced preliminary canopy delineation results, as illustrated in Figure 5, with most tree crowns accurately segmented and non-vegetative features excluded through canopy height model (CHM) filtering. However, residual artifacts—such as building windows exhibiting anomalous NDVI values—persisted due to spectral reflectance interference

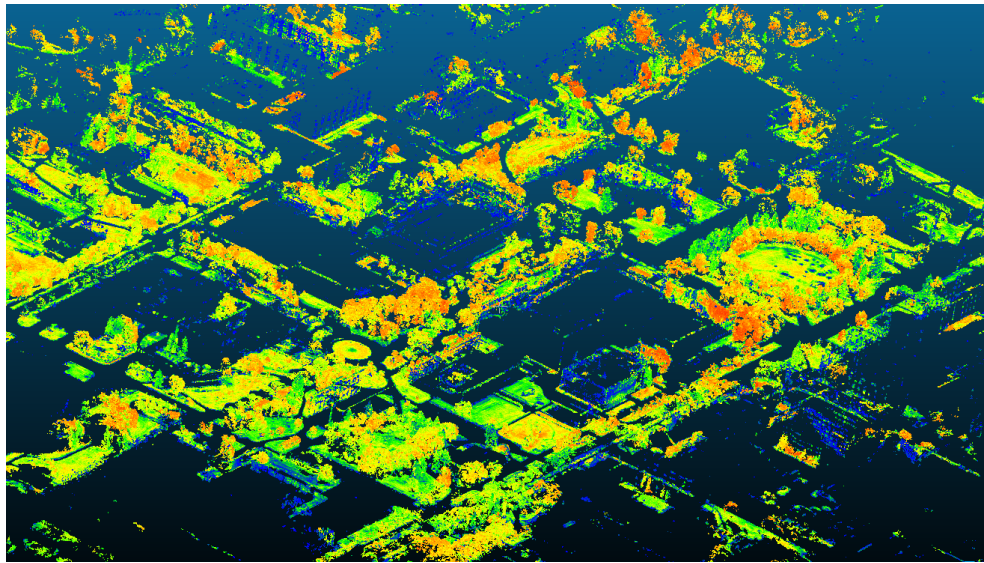


Figure 5. Tree crown segmentation using the Watershed method visualized in CloudCompare

To address this, post-delineation data cleaning was conducted. Polygons with areas below 2 m², indicative of non-tree objects (e.g., shrubs, artifacts), were removed, and roundness thresholds were applied to exclude irregularly shaped features. As demonstrated in Figure 6, these actions significantly improved segmentation accuracy, isolating tree crowns for downstream species classification.



Figure 6. The tree crowns after the delineation step visualized in QGIS.

It segmented and delineated trees both in tree dense areas (Figure 7) and in areas where trees are closed to buildings or even surrounded by them (Figure 8):

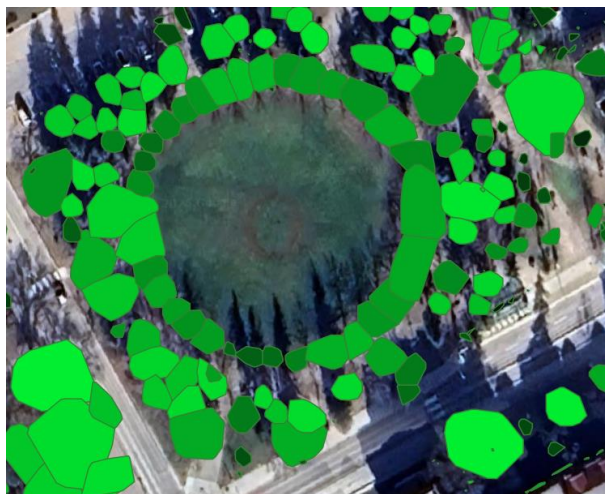


Figure 7. Tree dense area at the campus

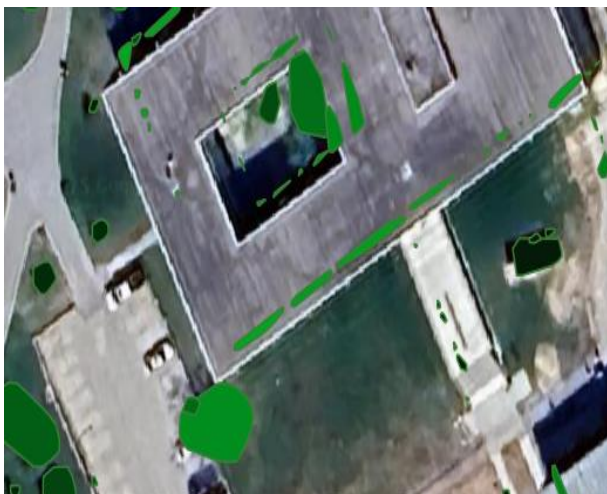


Figure 8. Trees around building

4.3 Labeling

Labeling was a difficult task as it is required to merge two different datasets. The first dataset contained the polygons created by the crown delineation. The second dataset contained the coordinates for the center of each tree and their species gathered by the University of Manitoba. The function “sf_join” in the “sf” package in R is used to merge these two datasets by checking if the coordinates of the second dataset fell inside of the polygons created previously. However, on multiple occasions the larger polygon contained multiple different species making that polygon unusable. After removing those larger unusable polygons, there were 817 polygons left that were properly labelled as seen below in figure 9.

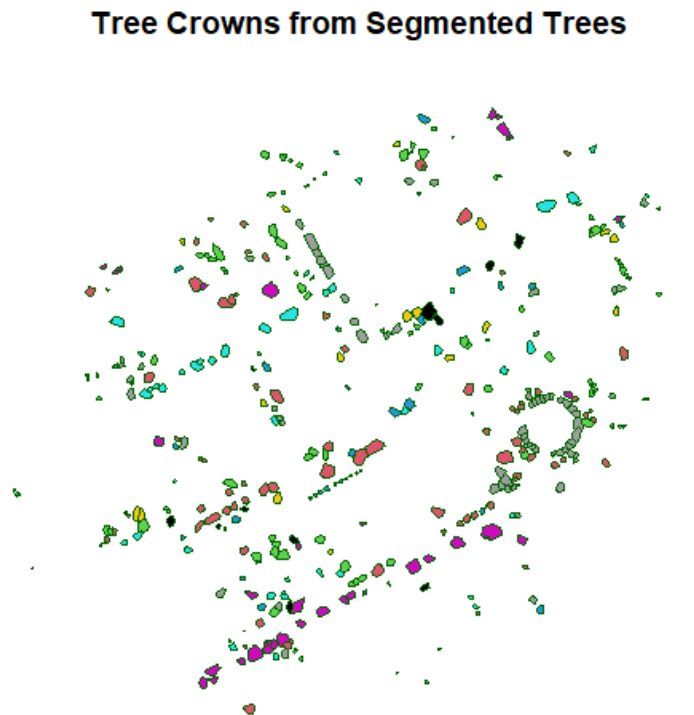


Figure 9. Geographic Distribution of Tree Species Across the Study Area

Note. Due to the high diversity of tree species ($n = 20$), the legend could not be feasibly incorporated within the figure.

4.4 Random Forest classification

Overall Statistics						
Accuracy : 0.8571						
95% CI : (0.8053, 0.8996)						
No Information Rate : 0.1558						
P-Value [Acc > NIR] : < 2.2e-16						
Kappa : 0.8427						
McNemar's Test P-Value : NA						
Statistics by Class:						
	Class: American Elm	Class: Amur Maple	Class: Ash-Mancana	Class: Basswood	Class: Burr Oak	Class: Cedar
Sensitivity	0.78571	0.9677	1.00000	0.8529	1.000000	1.00000
Specificity	0.98157	0.9950	1.00000	0.9645	0.995652	1.00000
Pos Pred Value	0.73333	0.9677	1.00000	0.8056	0.500000	1.00000
Neg Pred Value	0.98611	0.9950	1.00000	0.9744	1.000000	1.00000
Prevalence	0.06061	0.1342	0.02165	0.1472	0.004329	0.02165
Detection Rate	0.04762	0.1299	0.02165	0.1255	0.004329	0.02165
Detection Prevalence	0.06494	0.1342	0.02165	0.1558	0.008658	0.02165
Balanced Accuracy	0.88364	0.9814	1.00000	0.9087	0.997826	1.00000
	Class: Colorado Blue Spruce	Class: Crabapple-Rosybloom	Class: Elm-Japanese	Class: Golden Willow		
Sensitivity	0.71429	1.00000	0.333333	1.00000		
Specificity	0.99107	1.00000	1.000000	1.00000		
Pos Pred Value	0.71429	1.00000	1.000000	1.00000		
Neg Pred Value	0.99107	1.00000	0.991304	1.00000		
Prevalence	0.03030	0.01299	0.012987	0.06926		
Detection Rate	0.02165	0.01299	0.004329	0.06926		
Detection Prevalence	0.03030	0.01299	0.004329	0.06926		
Balanced Accuracy	0.85268	1.00000	0.666667	1.00000		
	Class: Green Ash	Class: Jack Pine	Class: Japanese Tree Lilac	Class: Manitoba Maple	Class: Scots Pine	
Sensitivity	0.8056	0.250000	1.00000	0.57143	0.92857	
Specificity	0.9538	0.982379	1.00000	1.00000	0.99539	
Pos Pred Value	0.7632	0.200000	1.00000	1.00000	0.92857	
Neg Pred Value	0.9637	0.986726	1.00000	0.98678	0.99539	
Prevalence	0.1558	0.017316	0.02165	0.03030	0.06061	
Detection Rate	0.1255	0.004329	0.02165	0.01732	0.05628	
Detection Prevalence	0.1645	0.021645	0.02165	0.01732	0.06061	
Balanced Accuracy	0.8797	0.616189	1.00000	0.78571	0.96198	
	Class: Siberian Crabapple	Class: Siberian Elm	Class: Silver Maple	Class: White Spruce	Class: White Willow	
Sensitivity	0.66667	0.87500	1.00000	0.666667	1.00000	
Specificity	1.00000	0.99535	1.00000	0.991228	0.99541	
Pos Pred Value	1.00000	0.93333	1.00000	0.500000	0.92857	
Neg Pred Value	0.98667	0.99074	1.00000	0.995595	1.00000	
Prevalence	0.03896	0.06926	0.02165	0.012987	0.05628	
Detection Rate	0.02597	0.06061	0.02165	0.008658	0.05628	
Detection Prevalence	0.02597	0.06494	0.02165	0.017316	0.06061	
Balanced Accuracy	0.83333	0.93517	1.00000	0.828947	0.99771	

Figure 10. Matrix of the accuracy for each species

When dealing with any kind of vegetation the most common classification method is the Random Forest classifier as it handles complex relationships between multiple characteristics efficiently. On the first pass through of the algorithm the labelled dataset was split 70/30 (training and testing sets respectively) and we got an accuracy score of 86%. The figure 10 above shows that most of the species were identified with high

accuracy. However, the species with a low amount of sample shown in figure 11 showed much lower accuracy therefore lowering the overall accuracy of the algorithm.

Species <fctr>	count <int>
American Elm	65
Amur Maple	94
Ash-Mancana	22
Basswood	118
Burr Oak	15
Cedar	11
Colorado Blue Spruce	33
Crabapple-Rosybloom	19
Elm-Japanese	16
Golden Willow	42

Figure 11. The number of samples from each tree species in the labelled dataset

The Random Forest algorithm also shows the importance of every feature and how they affect the accuracy of the model.

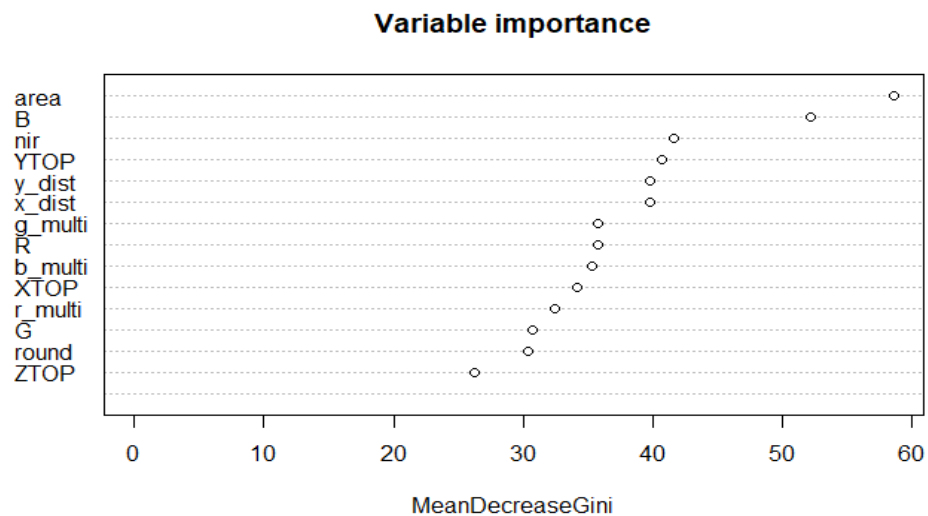


Figure 12. Top 15 importance features that characterize the dataset

The figure above (Fig.12) shows the importance of the top 15 features that characterize the data set. We can see the importance of area (Area of the canopy), the colour blue (from the RGB dataset), the height of the canopy (YTOP), and some of the multispectral colours (g_multi, b_multi and r_multi). We then ran the model again while removing the height of the canopy and received an accuracy of 78% which is a significant dip from the previous 86%. This showed the importance of having spatial information when classifying tree species.

To optimize the Random Forest, we did a grid search looking for the optimal depth for each decision tree and found that the optimal depth was 5. However, this did not increase the accuracy of the model implying that the bottleneck was most likely the sample size for some of the tree species.

5. Discussion

5.1 Contribution

The results show that supervised learning suits much better for the task of tree classification. While k-means clustering proved useful in the initial steps of classifying different types of objects (buildings, trees, ground, road), it was not useful for more detailed exploration of the point cloud.

Random Forests classifier, on the other hand, is efficient to achieve a high classification accuracy for the tree species. Our classification achieved a higher accuracy of 86%, outperforming the 83% reported by Hartling et al. (2021), who also applied a Random Forest model in an urban environment. However, it's important to mention that both these studies had some important differences, such the number of available tree species and tree samples.

Most importantly, we successfully achieved our objective of assessing the significance of the spatial characteristics of point cloud data in the context of tree canopy classification. This was shown by the height attribute, which improved our classification by 8%.

5.2. Limitations and future work

The integration of heterogeneous datasets—RGB and multispectral point clouds—presented a primary technical challenge, necessitating cross-platform workflows (e.g., Python-R) to reconcile spatial and spectral disparities.

A second challenge arose during the segmentation and delineation process. Although the watershed algorithm demonstrated robust performance in isolating individual tree crowns, its efficacy diminished in areas of high structural complexity, such as proximal canopies with overlapping crowns. Additionally, high NDVI reflectance from building windows led to the appearance of a few small unnecessary polygons. These limitations highlight the need for hybrid approaches combining spectral indices with contextual spatial rules to minimize false positives.

The classification accuracy of the Random Forest model was constrained by class imbalance, wherein species with limited training samples exhibited higher misclassification rates. However, we are confident this issue can be solved by simply adding more sample for this certain species.

Future research should prioritize the integration of deep learning architecture such as convolutional neural networks to develop a more robust and accurate classification model for similar tasks.

6. Conclusion

This study highlights the valuable role of data science, particularly machine learning, in addressing key tasks relevant to urban planning. The usage of UAS and 3D point cloud data proved highly effective, not only for accurate tree crown segmentation but also for classifying tree species with promising accuracy. The spatial characteristics of the point cloud data played a crucial role in the accuracy and effectiveness of the achieved results. We are confident that these results can be further improved by incorporating a larger number of tree samples and potentially integrating additional multispectral bands.

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